
Observability and the Fragility of Cooperation in Heterogeneous Multi-Agent Reinforcement Learning

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Abstract

We study how *observational heterogeneity*—differences in what agents perceive about a shared interaction—shapes cooperation in multi-agent reinforcement learning. Using collective reinforcement-learning dynamics (CRLD), we model two memory-two agents in an Iterated Prisoner’s Dilemma where one observes the full history and the other is selectively blind along controlled dimensions. Across more than a thousand initializations per scenario, cooperation is *fragile*: the deterministic dynamics mostly converge to mutual defection, and every restriction of the partial observer reduces cooperation. The collapse is sharpest—essentially total—when an agent cannot perceive its *own* past actions, the substrate reciprocity requires. The two agents nonetheless earn nearly identical reward (the better-informed agent gains no systematic advantage; exploitation is rare), so the dominant failure mode of heterogeneity is shared defection, not domination. Reading the dynamics as a flow in policy space makes these basins explicit, and a memory sweep shows the cooperative basin is itself created by memory—absent at memory one, it emerges and enlarges with history, a deterministic echo of the intrinsic-fluctuation route to cooperation. These corrected results temper earlier optimistic claims and isolate self-referential observability as a mechanism learning agents need to sustain reciprocity.

1 Introduction

Cooperation underlies much of human flourishing Hare and Woods [2020], Bar-Yam [2000], yet the complexity of modern societies routinely produces social dilemmas in which individual incentives and collective welfare diverge—from climate policy Kersting [2017] and pandemic response Glennerster et al. [2024] to political stalemate Frieden and Lake [2015], Dafoe et al. [2020] and economic distrust Acemoglu and Robinson [2012]. Such problems are increasingly modeled with Markov Decision Processes and reinforcement learning (RL), which capture how agents—governments, firms, consumers—adapt strategies from partial observations of an evolving environment Glennerster et al. [2024], Charpentier et al. [2021].

A defining feature of real-world actors is that they do *not* perceive their shared environment identically: a firm and a regulator see different slices of the same market, and treaty negotiators hold asymmetric information about each other—and about their own commitments. We therefore ask a focused question: *how does heterogeneity in observability shape whether learning agents come to cooperate?* We isolate observability from other heterogeneities (learning rate, risk tolerance) so its effect can be measured cleanly, in the general-sum Iterated Prisoner’s Dilemma Barfuss [2022].

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This paper revisits and corrects an earlier work-in-progress study by the present authors, whose preliminary report suggested that cooperation between knowledge-asymmetric agents emerges readily. A careful, parameter-controlled reproduction shows a more nuanced picture: cooperation is achievable but *fragile*, most observational impairments suppress it, and the loss of *self*-observability is uniquely destructive. We present the corrected methodology, equations, and regenerated results.

2 Methods

2.1 Learning dynamics

We simulate two agents, $i \in \{1, 2\}$, each choosing an action $a^i \in \{c, d\}$. Rather than noisy, sample-based simulations, we use *collective reinforcement learning dynamics* (CRLD) Barfuss [2023, 2022], Barfuss et al. [2019], the deterministic, infinite-batch (strategy-averaged) limit of temporal-difference learning: it replaces stochastic value updates with deterministic equations in *strategy space* that are efficient and analytically transparent. Each agent holds a policy $x^i(a | o)$ over actions given an *observation* o , which need not coincide with the true state s —the lever for perceptual heterogeneity (Section 2.3). The policy updates along the strategy-averaged temporal-difference error δ^i via the policy-space replicator–mutator map Barfuss [2022], Barfuss et al. [2019] (not a tabular value update):

$$x_{t+1}^i(a | o) = \frac{x_t^i(a | o) \exp(\alpha_i \beta_i \delta_t^i(o, a))}{\sum_b x_t^i(b | o) \exp(\alpha_i \beta_i \delta_t^i(o, b))}, \quad (1)$$

where $\alpha_i \in (0, 1]$ is the learning rate and $\beta_i > 0$ is the choice intensity (inverse temperature) controlling the exploration–exploitation trade-off. Equation (1) replaces the explicit ϵ -greedy action selection and tabular SARSA value update of standard RL; in the deterministic limit, softmax action selection with intensity β_i is folded directly into the policy and its update.

The strategy-averaged temporal-difference error for actor–critic learning is

$$\delta_t^i(o, a) = (1 - \gamma_i) \bar{R}^i(o, a) + \gamma_i \bar{V}_{\text{next}}^i(o, a) - \bar{V}^i(o), \quad (2)$$

where $\gamma_i \in [0, 1)$ is the discount factor, $\bar{R}^i(o, a)$ the strategy-averaged immediate reward, $\bar{V}^i(o)$ the value of observation o under the joint policy, and $\bar{V}_{\text{next}}^i(o, a)$ the strategy-averaged successor value; the $(1 - \gamma_i)$ prefactor puts immediate and future returns on the same scale Barfuss et al. [2019]. Using the expected successor value makes this the deterministic, on-policy form of TD learning (the infinite-batch limit of Expected SARSA Sutton and Barto [2018]); the values $\bar{V}^i$ solve the corresponding Bellman system in closed form. Under partial observability all averages additionally integrate the belief over true states induced by the observation model, following the partially observable CRLD formulation Barfuss and Mann [2022].

The dynamics as a flow. Equations (1)–(2) define a deterministic map on the joint policy—equivalently, a vector field (a *flow*) on strategy space. From any joint policy the learning rule points in a definite direction, the fixed points where the flow vanishes are the learned equilibria, and the *cooperation rate* we report is precisely the measure of the basin of attraction of the cooperative fixed point under random initial policies. Visualizing this flow (Figures 1–3) exposes *why* a scenario does or does not cooperate, complementing the reward statistics with the geometry behind them.

Because α_i , β_i , and γ_i can be set per agent, this framework natively supports *heterogeneous* agents that differ in learning speed and effective risk tolerance. In this paper we hold these parameters equal across agents (Appendix) so that the sole heterogeneity under study is observability.

2.2 Environment

The environment is the Iterated Prisoner’s Dilemma (IPD). The one-shot payoffs (Table 1) are $R = 1$ (mutual cooperation), temptation $T = 1.2$, sucker $S = -0.5$, and punishment $P = 0$, satisfying $T > R > P > S$ and $2R > T + S$. We embed the stage game into a *memory-two* history (an agent’s state is the previous two rounds’ outcomes)—the smallest history rich enough to host the reciprocal

strategies (win-stay-lose-shift, tit-for-tat) that direct reciprocity relies on Schmid et al. [2021], Hilbe et al. [2017], yet low-dimensional enough for exhaustive analysis.

Table 1: Payoff matrix for the Iterated Prisoner’s Dilemma (row agent, column agent).

	c	d
c	(1, 1)	(-0.5, 1.2)
d	(1.2, -0.5)	(0, 0)

2.3 Observational heterogeneity

Agent 1 is a *full observer* (perceives both agents’ last two actions without aliasing). Agent 2 is a *partial observer*: along controlled dimensions, parts of the most recent round are masked, aliasing several true states to one observation (its belief is spread uniformly over them) while the prior round stays fully observed. We study four restrictions of Agent 2; exact state-to-observation maps are in the Appendix (Tables 2–5).

- **Self-aware:** Agent 2 remembers its *own* most recent action but not Agent 1’s (it cannot see what the other did last round).
- **Non-self-aware:** Agent 2 remembers Agent 1’s most recent action but not its *own* (it cannot see what it itself did last round).
- **Cooperation-tracking:** Agent 2 perceives Agent 1’s last action only when Agent 1 *cooperated*; Agent 1’s defections are masked.
- **Defection-tracking:** Agent 2 perceives Agent 1’s last action only when Agent 1 *defected*; Agent 1’s cooperations are masked.

For each scenario we run the deterministic dynamics from $N = 1,200$ random softmax initial policies until a fixed point is reached (tolerance 10^{-5} , at most 4,000 steps), discard the small fraction of non-converged runs, and record each agent’s converged average reward $\bar{R}^i$. We summarize outcomes by the rate of *mutual cooperation* (both agents’ average reward > 0.9 , i.e. close to $R = 1$) and by mean reward per agent.

3 Results

We read the same dynamics in two complementary ways: as the flow geometry in policy space (Figure 1), and as reward statistics over $N = 1,200$ random initializations per scenario (Figure 2).

Cooperation is fragile, and impairing observability suppresses it. Under full observability the dynamics reach mutual cooperation in only about 14% of random initializations; the rest collapse to defection. The flow makes this concrete: even the full-observability panel of Figure 1 has most arrows pointing into the defection corner, with only a small basin draining to $(1, 1)$ —so even with perfect, symmetric information, defection is the modal attractor at these parameters. Every observational restriction of Agent 2 shrinks the cooperative basin further (Figures 1, 2a), making observational completeness a prerequisite—not a guarantee—for cooperation.

Losing self-observability is catastrophic. The starkest contrast concerns *whose* action is hidden. When Agent 2 cannot observe its *own* last action (non-self-aware), the flow drains uniformly to defection (Figure 1, top-right) and essentially every run converges to mutual defection, with both agents earning a reward indistinguishable from the punishment payoff $P = 0$. Retaining self-observability while losing sight of the *other* agent (self-aware) is also damaging but strictly less so: a nonzero fraction of trajectories still climb toward cooperation (Figure 1, top-middle). Mechanistically, an agent that cannot identify its own move cannot implement a reciprocal, history-dependent rule at all—it cannot “stay” or “shift” relative to its own behavior—so the substrate for direct reciprocity disappears. This recovers and sharpens the earlier observation that the absence of self-awareness is uniquely destructive, while clarifying that self-awareness is necessary but far from sufficient.

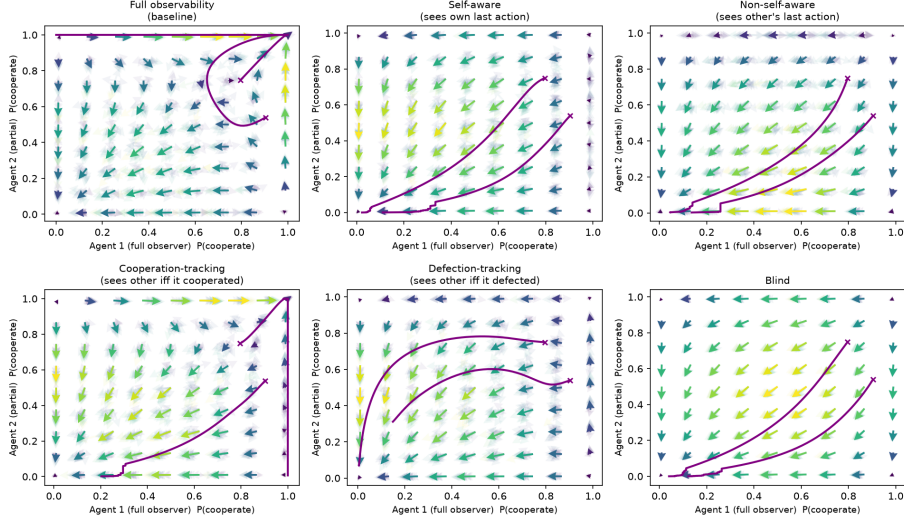


Figure 1: Learning flow of the heterogeneous memory-two IPD (actor-critic CRLD), one panel per observability condition of the partial observer (Agent 2; Agent 1 is always a full observer). Arrows are the deterministic learning update at each joint policy; purple curves are two trajectories; axes are each agent’s cooperation probability. Mutual cooperation is the corner $(1, 1)$, defection $(0, 0)$. A basin draining to cooperation exists under full observability, cooperation-tracking, and—weakly—self-awareness, but the flow collapses toward defection when the partial observer cannot see its *own* last action (non-self-aware) or is blind.

What an agent chooses to track matters, but exploitation is rare. Conditioning *which* of the other’s moves are visible also shapes outcomes. An agent that perceives the opponent’s cooperations (cooperation-tracking) sustains more cooperation than one that perceives only its defections (defection-tracking); see the corresponding panels of Figure 1 and Figure 2a. Visibility of the partner’s cooperation provides the positive signal on which reciprocity can build, whereas a perception filtered toward the partner’s defections biases the learner toward distrust. Importantly, we do *not* observe systematic exploitation in either case: the joint-reward distributions are close to symmetric, and one-sided outcomes (one agent at the temptation payoff, the other at the sucker payoff) are uncommon. The earlier framing of these scenarios as “Agent 2 being exploited” versus “exploiting” is not supported; the dominant effect of perceptual filtering is on the *shared* cooperation rate, not the division of spoils.

Memory creates and enlarges the cooperative basin. The cooperative basin is not a fixed feature of the game but a product of *memory*. Re-running the full-observability dynamics at memory one, two, and four (Figure 3) shows that at memory one the learning flow does not reach mutual cooperation from our initializations; a cooperative fixed point at $(1, 1)$ becomes reachable only at memory two, and its basin enlarges at memory four. Longer histories let agents represent richer reciprocal strategies (memory- k tit-for-tat / win-stay-lose-shift) that turn cooperation into a deterministic attractor. This is the deterministic counterpart of the finite-batch *intrinsic fluctuations* that promote cooperation in the memory-one IPD Barfuss and Meylahn [2023] (Appendix, Figure 3): memory and sampling noise enlarge the cooperative basin by different routes. It also justifies memory two as the smallest setting in which heterogeneous reciprocity can be studied at all.

Heterogeneity costs both agents, and favors neither. The two agents earn almost identical average reward in every scenario (Figure 2b): the gap is below 0.015 in reward units everywhere, it is not consistently in favor of the better-informed agent, and in the cooperation-tracking scenario the *partial* observer is in fact marginally ahead. The better-informed agent enjoys no systematic advantage from its superior perception. This contradicts our earlier report that more knowledgeable agents reliably earn more; we trace that claim to a parameter bug (Appendix), and find that with a valid discount factor the apparent advantage disappears or reverses. The headline of heterogeneity is therefore mutual loss—both agents forgo the cooperative surplus—rather than a winner taking the other’s share.

The aggregate cooperation rates and per-agent rewards across all scenarios are summarized in the Appendix (Figure 2); a representative trajectory (Figure 4) corroborates this picture.

4 Discussion

The corrected results support a cautionary reading of heterogeneous-observability cooperation. In the deterministic learning limit, a heterogeneous Iterated Prisoner’s Dilemma does not spontaneously favor cooperation; it favors defection, and perceptual asymmetries make matters worse rather than better. The clearest positive statement we can make is structural: an agent must be able to observe its own past behavior to participate in reciprocity at all, and shared visibility of cooperative moves is what allows the residual cooperation to persist. Where our earlier preliminary report read these simulations optimistically, the controlled reproduction reads them as a map of *failure modes*—which perceptual deficits are survivable and which are fatal.

We note that other lines of work recover robust cooperation in closely related settings by departing from the strict deterministic limit—most relevantly, intrinsic fluctuations from finite-batch sampling can stabilize cooperation that the infinite-batch dynamics miss Barfuss and Meylahn [2023]. Our memory result (Figure 3) is a complementary, purely deterministic route to the same place: enlarging the agents’ history enlarges the cooperative basin without invoking noise at all. That both extra memory and finite-batch fluctuations expand the same basin suggests cooperation here is gated by how much reciprocal structure the learners can represent and stabilize; reconciling these mechanisms under the observability deficits studied here—in particular whether fluctuations can rescue the non-self-aware case that memory alone cannot—is a natural next step.

5 Future work and limitations

Natural next steps are to (i) sweep β and γ to map where, if anywhere, observational heterogeneity becomes cooperation-promoting; (ii) reintroduce finite-batch fluctuations Barfuss and Meylahn [2023] to test whether they rescue cooperation under partial observability where memory alone cannot; and (iii) extend beyond the IPD to public-goods, bargaining, and common-pool games, eventually connecting these low-dimensional dynamics to deep-RL and language-model agents. The present conclusions are correspondingly bounded: the IPD captures a narrow slice of strategic behavior; CRLD assumes deterministic, infinite-batch update rules rather than human idiosyncrasy; outcomes depend on the chosen parameters and on the single regime we report; and cooperation rates are statements about the measure of a basin of attraction under random initial policies, not about any single trajectory.

6 Acknowledgments

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A Appendix / supplemental material

A.1 Software and reproducibility

All experiments use an extension of the open-source pyCRLD library Barfuss [2023] that adds heterogeneous per-agent observation tensors. The deterministic dynamics, observation-tensor construction, and figure-generation scripts used for every result and figure in this paper are released with the camera-ready version.

Corrections relative to the earlier preliminary report. The present version fixes four issues in the earlier exploratory code and writeup. (1) *Invalid discount factor*: the mixed-observability scenario underlying the earlier “more knowledgeable agents earn more” claim had been run with a discount factor of 10, outside the valid range $[0, 1]$. This makes the $(1 - \gamma)$ prefactor negative and the value-function solve ill-posed; in a controlled comparison it both inflates the apparent cooperation rate (here, from $\approx 14\%$ to $\approx 21\%$) and manufactures a spurious reward advantage for the full observer that vanishes—indeed reverses—at $\gamma = 0.9$. All scenarios here use $\gamma = 0.9$. (2) *Stale-variable reuse*: several cells reported cooperation percentages from a variable computed for the *previous* scenario, so some printed statistics did not correspond to the figure beside them; and some figure legends carried hard-coded percentages rather than values derived from the data. (3) *Math*: the learning rule is now stated as the policy-space CRLD update of Eqs. (1)–(2) actually implemented by the framework, rather than a tabular SARSA value update. (4) *Consistent pipeline*: all cooperation and exploitation statistics are re-derived under one identical, controlled pipeline across all scenarios, which removes per-scenario drift and overturns the earlier exploitation narrative.

A.2 Parameters

For all experiments we use learning rate $\alpha = 0.1$, discount factor $\gamma = 0.9$, and choice intensity $\beta = 1$, identical across both agents. Memory length is two ($h = (2, 2, 2)$ in the history-embedding convention). Each scenario uses $N = 1,200$ random softmax initial policies; trajectories run up to 4,000 steps with fixed-point tolerance 10^{-5} .

A.3 Aggregate statistics and the cooperative basin

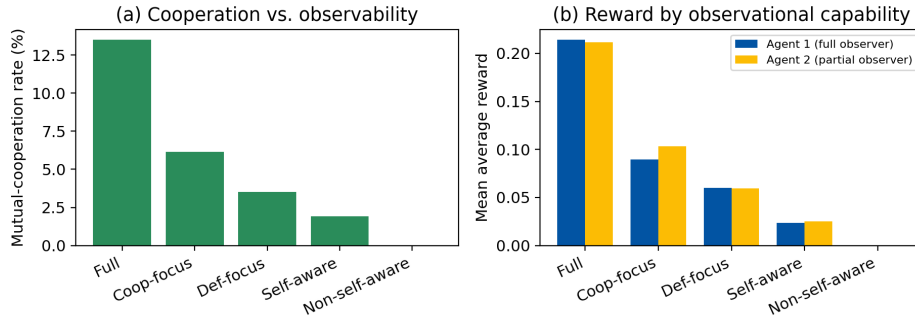


Figure 2: Summary across observability scenarios, $N = 1,200$ random initial conditions each. (a) Mutual-cooperation rate decreases monotonically as Agent 2’s observability is degraded, from full observability down to the non-self-aware case. (b) Mean average reward per agent; the full observer (Agent 1) is never meaningfully worse off than the partial observer (Agent 2), but the gap is small and both fall far short of the cooperative payoff $R = 1$.

A.4 Representative trajectory

A.5 Observation maps

Tables 2–5 give the state-to-observation maps for Agent 2 in each scenario. Each row is one of the 16 memory-two states; entries show what Agent 2 perceives, with * denoting a masked (unobserved)

Full observability: the cooperative basin emerges and grows with memory (IPD, actor-critic)

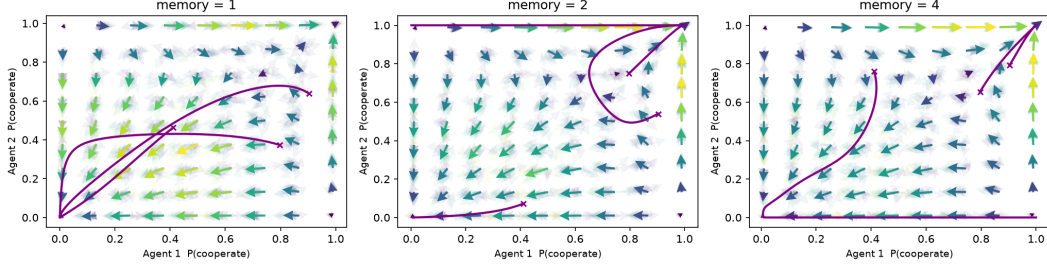


Figure 3: Full-observability IPD learning flow at memory one, two, and four (actor-critic). A cooperative fixed point at $(1, 1)$ is not reached at memory one but becomes an attractor from memory two onward, and its basin enlarges with memory—a deterministic analogue of the intrinsic-fluctuation mechanism Barfuss and Meylahn [2023].

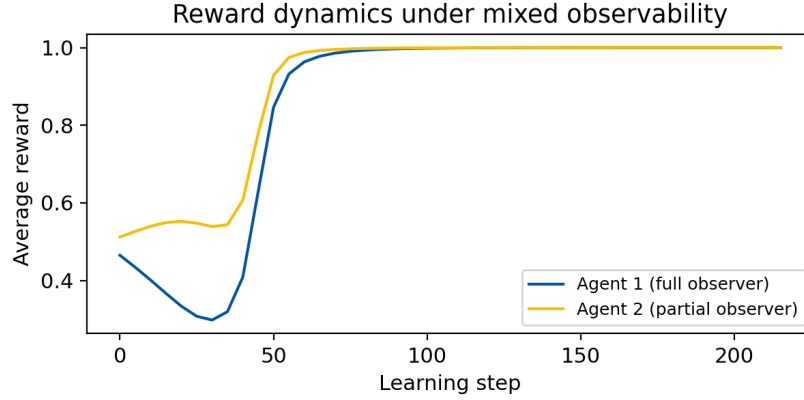


Figure 4: A representative learning trajectory under mixed observability. Average reward per agent versus learning step; the agents settle near the defection payoff, consistent with the aggregate finding that defection is the modal attractor.

coordinate that aliases the state with others sharing the same observation. Agent 1 observes every state without aliasing in all scenarios.

Table 2: Self-aware Agent 2: keeps its own most recent action, masks Agent 1’s.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(*, C)
(C, C)	(C, D)	(C, C)	(*, D)
(C, D)	(C, C)	(C, D)	(*, C)
(C, D)	(C, D)	(C, D)	(*, D)
(D, C)	(D, C)	(D, C)	(*, C)
(D, C)	(D, D)	(D, C)	(*, D)
(D, D)	(D, C)	(D, D)	(*, C)
(D, D)	(D, D)	(D, D)	(*, D)

Table 3: Non-self-aware Agent 2: keeps Agent 1's most recent action, masks its own.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(C, *)
(C, C)	(C, D)	(C, C)	(C, *)
(C, D)	(C, C)	(C, D)	(C, *)
(C, D)	(C, D)	(C, D)	(C, *)
(D, C)	(D, C)	(D, C)	(D, *)
(D, C)	(D, D)	(D, C)	(D, *)
(D, D)	(D, C)	(D, D)	(D, *)
(D, D)	(D, D)	(D, D)	(D, *)

Table 4: Cooperation-tracking Agent 2: sees Agent 1's last action only when Agent 1 cooperated.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(C, C)
(C, C)	(C, D)	(C, C)	(C, D)
(C, C)	(D, C)	(C, C)	(*, C)
(C, C)	(D, D)	(C, C)	(*, D)
(D, C)	(C, C)	(D, C)	(C, C)
(D, C)	(C, D)	(D, C)	(C, D)
(D, C)	(D, C)	(D, C)	(*, C)
(D, C)	(D, D)	(D, C)	(*, D)

Table 5: Defection-tracking Agent 2: sees Agent 1's last action only when Agent 1 defected.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(*, C)
(C, C)	(C, D)	(C, C)	(*, D)
(C, C)	(D, C)	(C, C)	(D, C)
(C, C)	(D, D)	(C, C)	(D, D)
(D, C)	(C, C)	(D, C)	(*, C)
(D, C)	(C, D)	(D, C)	(*, D)
(D, C)	(D, C)	(D, C)	(D, C)
(D, C)	(D, D)	(D, C)	(D, D)